Electric Fields

17.1 Concept of an electric field

17.2 Uniform electric fields

17.3 Force between point charges

17.4 Electric field of a point charge

17.5 Electric potential

$$V = \frac{Q}{4\pi\varepsilon_o r}$$

$$k = \frac{1}{4\pi\varepsilon_0}$$

$$E = \frac{kq}{r^2}$$

$$F = k\frac{q_1 q_2}{r^2}$$

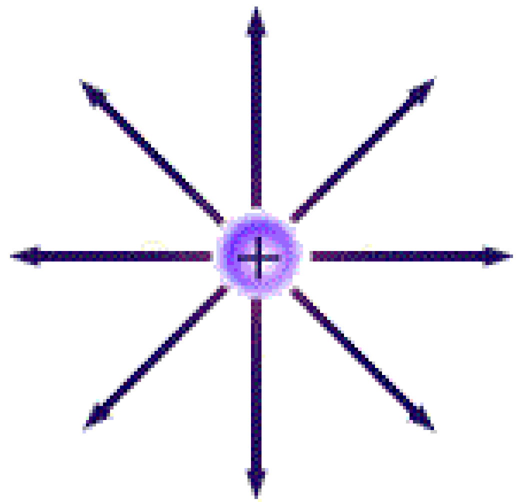
$$k = 8.99 \times 10^9 \frac{\mathrm{N\,m^2}}{\mathrm{C^2}}$$

Electric Fields

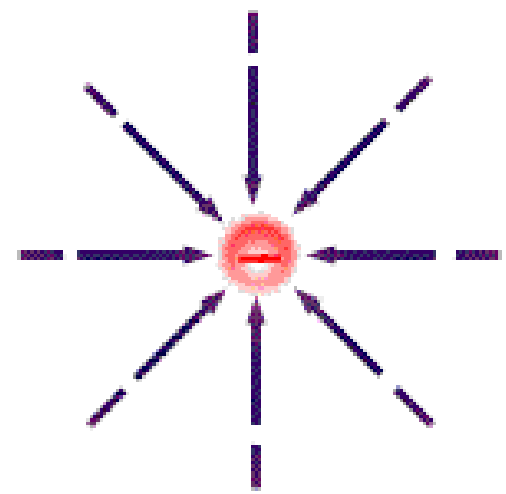
- Electric charges exert forces on each other when they are a distance apart. The word ‘Electric field’ is used to explain this action at a distance.
- An Electric field is defined as the region of space where a stationary charge experiences force.

The direction of Electric Fields

- The direction of electric field is defined as the direction in which a positive charge would move if it were free to do so. So the lines of force can be drawn with arrows that go from positive to negative.
- Electric field lines are also called force lines.
- The field lines are originated from the positive charge and they end up at the negative charge.



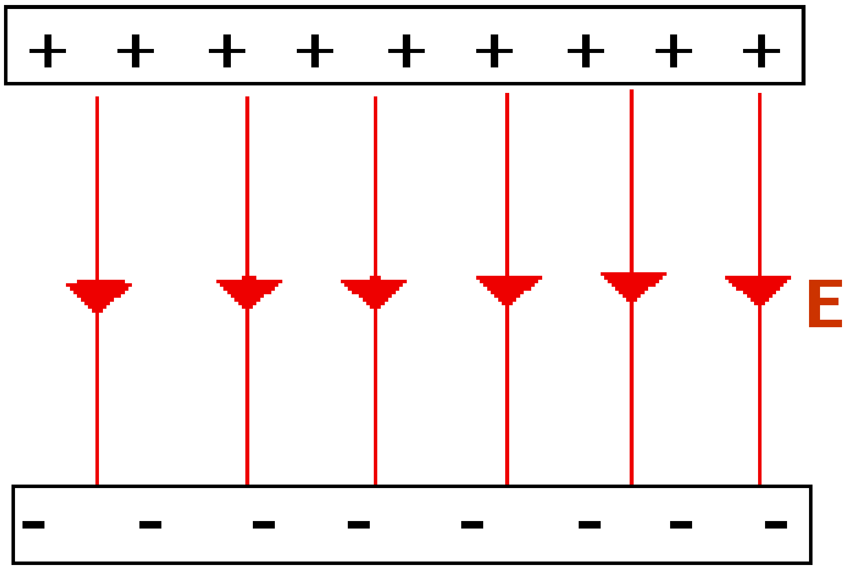
Positive Charge Electric Field
(animated demo)



Negative Charge Electric Field
(animated demo)

Remember, for any ELECTRIC FIELD.....

- The lines of force starts on a positive charge and end on a negative charge.
- The lines of force never touch or cross.
- The strength of the electric field is indicated by the closeness of the lines; means the closer they are, the stronger the field.



Electric Field Strength

- Electric field strength at a point is defined as the force per unit charge acting on a small positive charge placed at that point.
- If a force experienced by a positive charge +Q placed in the field is F, then the field strength, E is given by $E = \frac{F}{Q}$

Note : Remember, the symbol E is also used for 'energy'.

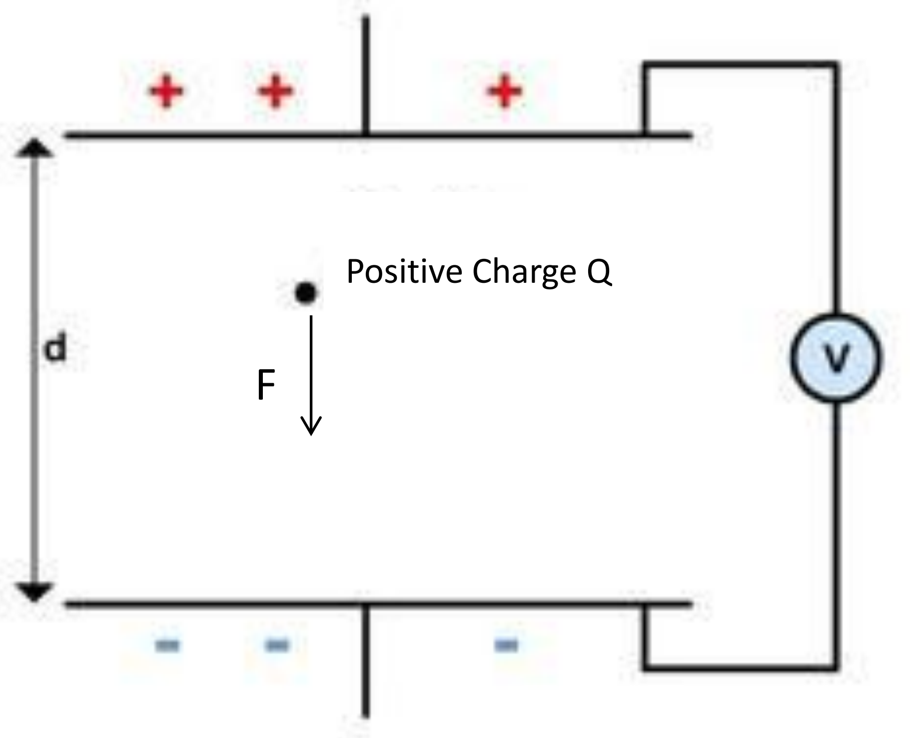
Unit of Electric Field Strength

- The unit of Electric field strength for the equation $E = \frac{F}{Q}$ is given by N C^{-1}

where force is measured in Newtons and charge in Coulombs.

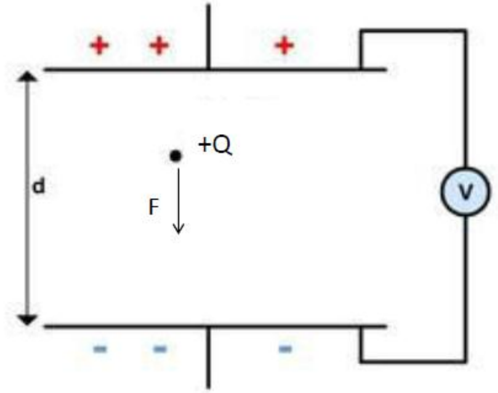
Note : Remember, later we shall see that there is another common SI unit for electric field strength, Vm^{-1}. However the 2 units are equivalent.

The field strength (E) of the uniform field between charged parallel plates in terms of potential difference (V) and separation (d)



The field strength of the uniform field between charged parallel plates in terms of potential difference and separation.

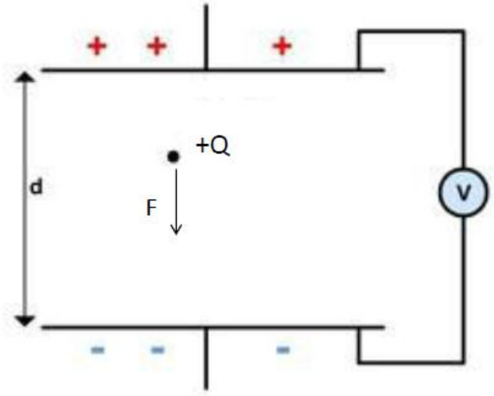
- The figure illustrates parallel plates at a distance d apart with a potential difference *V* between them.



- A charge +Q in the uniform field between the plates has a force *F* acting on it.
- To move the charge towards the positive plate would require work to be done on the charge.

Continued from previous slide…

- Work done is given by the product of force and distance.
- To move the charge from one plate to other requires work W and is given by $W = Fd$, where F is force and d is the distance.



- Now lets see what is potential difference. (see next slide)

Continued from previous slide…

So what is potential difference?

- If the electric field is ***NOT UNIFORM***, it is not so simple to calculate the energy change due to moving a charge in the field.

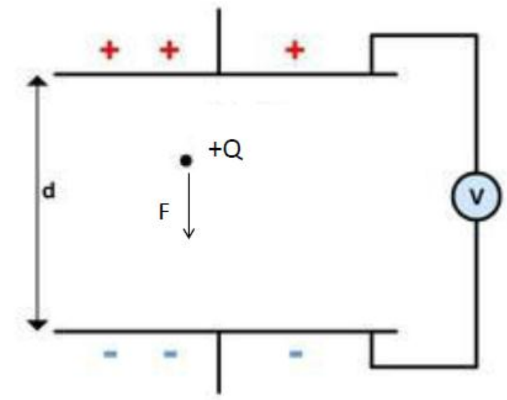
- *It is therefore useful to define a quantity which describes the work done in moving unit charge from one point in the field to another point*. We call this quantity the ***POTENTIAL DIFFERENCE*** between the two points and is given by $V = \frac{W}{Q}$

- V is the symbol for potential difference, which has units of ***JOULES PER COULOMB, (JC^{-1})***. As this is an important quantity, it is given its own unit, the ***VOLT, (V)***.

- ***"One volt is the potential difference between two points in an electric field such that one joule of work is done in moving one coulomb of charge from one point to another."***

Continued from previous slide…

- By rearranging equation on potential difference, work done W = *VQ*



- Thus W = Fd = *VQ*
- $$\frac{F}{Q} = \frac{V}{d}$$
- But $\frac{F}{Q}$ is the force per unit charge and this is the definition of electric field strength.
- Thus, for a uniform field, the field strength E is given by E = $\frac{V}{d}$

Continued from previous slide…

For parallel plates separated by a distance *d*, with a potential difference ΔV, the uniform electric field within the plates has strength:

$$E = \frac{\Delta V}{d}$$

We sometimes use an alternate unit for *E.*

$$E = \frac{Voltage}{\text{distance}}$$

$$= \frac{Volts}{metre}$$

$$= Vm^{-1}$$

Calculate the forces on charges in uniform electric fields.

$$\frac{F}{Q} = \frac{V}{d}$$

$$E = \frac{V}{d}, E = \frac{\Delta V}{d}$$

Sample problem 1 : Calculate E

Two parallel plates separated by **0.1m** have a potential difference **ΔV = 100V**. What is the Electric Field strength between the plates?

$$E = \frac{\Delta V}{d}$$

$$= \frac{100V}{0.1m}$$

$$= 1000Vm^{-1}$$

Sample problem 2 : Calculate E and F

- Two metal plates 5.0cm apart have a potential difference of 1000 V between them. Calculate :

(a) The strength of the electric field between plates.

(b) The force on a charge of 5.0 nC between the plates.

Solution :

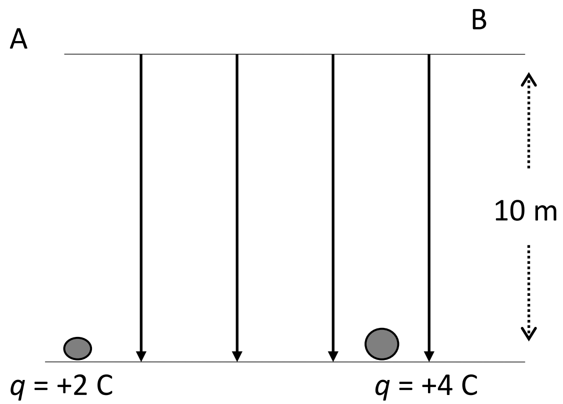
(a) E = $\frac{V}{d}$, E = $\frac{1000}{0.05}$ = $\underline{2.0 \times 10^{4}\ Vm^{-1}}$

(b) F = EQ, F = $2.0 \times 10^{4} \times 5.0 \times 10^{-9}$ = $\underline{1.0 \times 10^{-4}\ \text{N}}$

The effect of a uniform electric field on the motion of charged particles

Energy Changes in Electric Fields

Consider the movement of a charge in a uniform electric field ;



$\boldsymbol{E}$ = 10 N C^{-1}

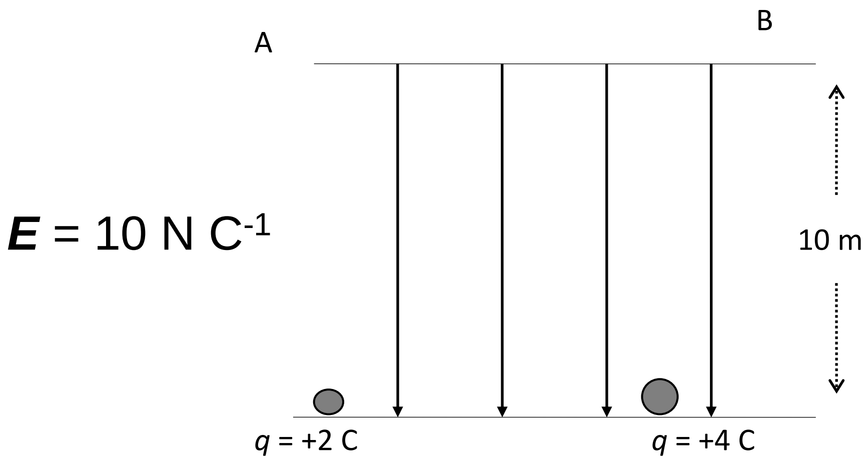
To lift a charge towards the top (positive) plate we exert an external force;
Therefore, Work Done by external Force is :

$$W = F_{ext} \times d$$
$$= EQ.d \qquad (\text{since } F = EQ)$$
$$= \frac{V}{d} \times Q \times d \qquad (\text{since } E = \frac{V}{d})$$

$$W = QV$$

Sample problem 3 : Calculate the Work done

The work done on each charges are :



$$w = QEd$$
$$= 2 \times 10 \times 10$$
$$= 200J$$

$$w = QEd$$
$$= 4 \times 10 \times 10$$
$$= 400J$$

Sample problem 3 : Calculate V

We **define** the **Work done** (in moving charge from one position to another) **per unit charge** as the change in potential or **potential difference, V.**

$$V = \frac{W}{q}$$

Taking the data of W from previous sample problem, calculate V.

+2C Charge,

$$V = \frac{200J}{2C}$$
$$= 100JC^{-1}$$

+4C Charge,

$$V = \frac{400J}{4C}$$
$$= 100JC^{-1}$$

Electron Volt

Work done when a charge of one electron moves through a potential difference of 1 V is one electron volt (e.V).

The equivalent energy is:

Since $W = QV$

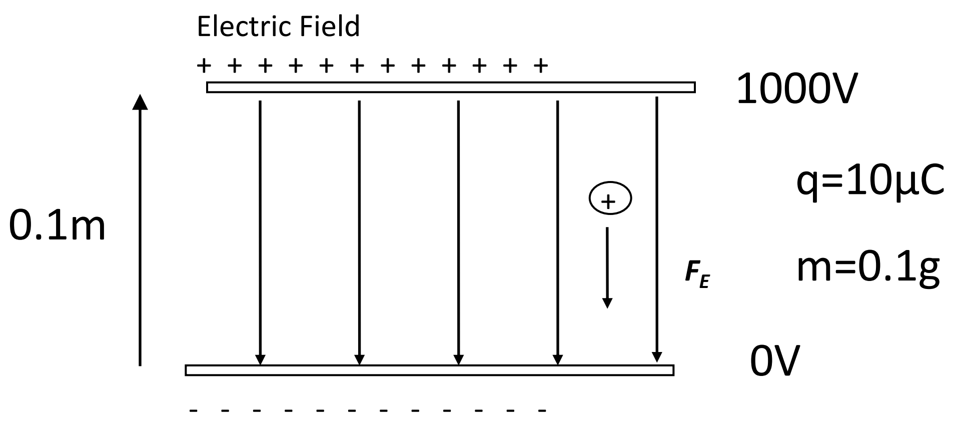
$\boldsymbol{Q}$ **= 1.6 x $\mathbf{10^{-19}}$C**, and **1 V = 1 J $\mathbf{C^{-1}}$**

hence 1 eV = 1.6 x 10^{-19}C x 1 J C^{-1}

1 eV = 1.6 x $\mathbf{10^{-19}}$ J

Motion in an Electric Field
Find the velocity using Equations of Motion
(with sample problem)

Consider a positive charge placed in a uniform electric field, as shown in the diagram below.



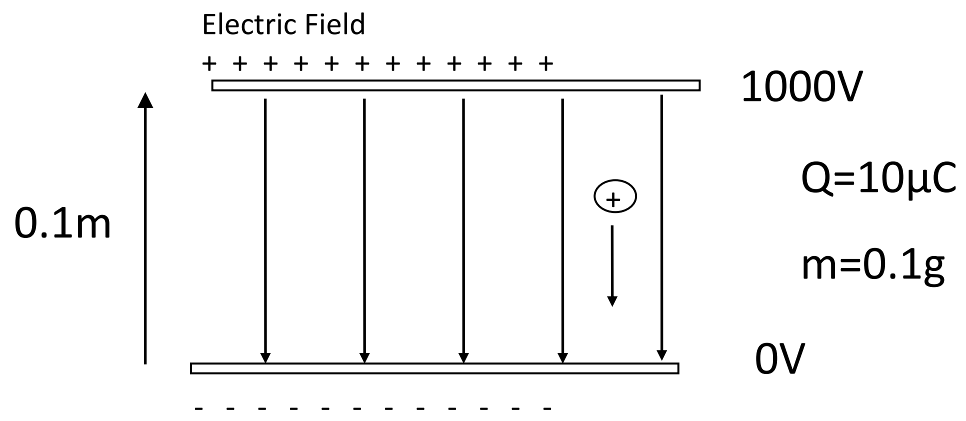
Find the velocity of the charge after it has travelled a distance of 5 cm. Use the following information:

Motion in an Electric Field

Find the velocity using Equations of Motion

(with sample problem)

(continued from previous slide)



$$E=\frac{V}{d}$$

$$=\frac{1000}{0.1}=1\times10^{4}Vm^{-1}$$

$$=1\times10^{4}NC^{-1}$$

$$F=QE$$

$$=10\times10^{-6}\times1\times10^{4}$$

$$=10^{-1}N$$

$$a=\frac{F}{m}$$

$$=\frac{10^{-1}}{10^{-4}}=10^{3}ms^{-2}$$

Motion in an Electric Field
Find the velocity using Equations of Motion
(with sample problem)
(continued from previous slide)

Can use the equations of motion to determine the speed of particle after travelling for 5cm.

$$v_1 = 0ms^{-1} \quad d = 0.05m \quad a = 10^3 ms^{-2} \quad v_2 = ?$$

$$v_2^2 - v_1^2 = 2as$$

$$v_2^2 = 2as \quad (v_1 = 0ms^{-1})$$

$$v_2 = \sqrt{2as}$$

$$v_2 = \sqrt{2 \times 10^3 \times 0.05}$$

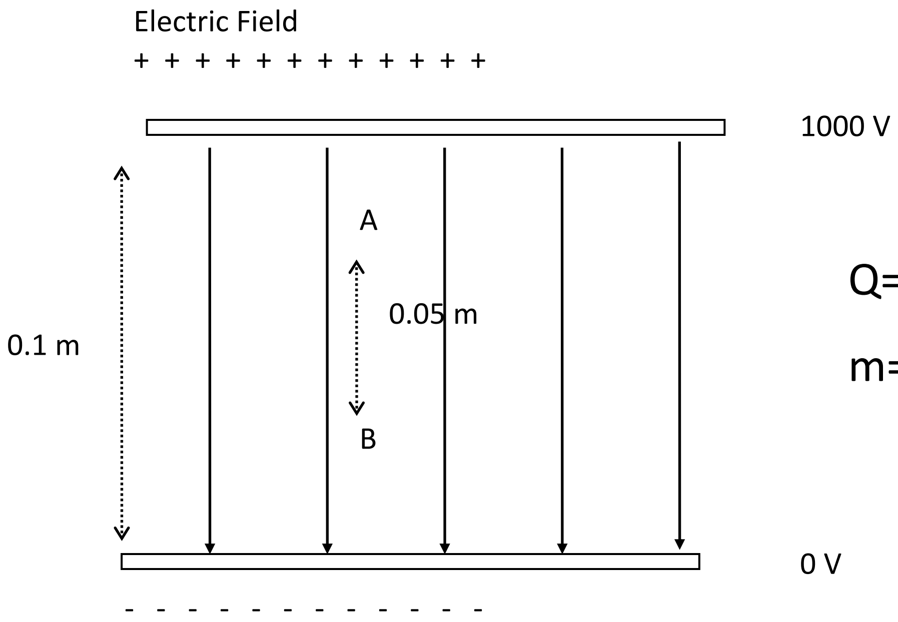
$$v_2 = 10ms^{-1} \text{ towards the -'ve plate}$$

Motion in an Electric Field
Find the velocity using Change in K.E
(with sample problem)

Can also determine the velocity by using the change in kinetic energy of the particle.

$$E = \frac{\Delta V}{d}$$

$$= 10000 Vm^{-1}$$



Q=10μC

m=0.1g

Motion in an Electric Field

Find the velocity using Change in K.E

(with sample problem)

(Continued)

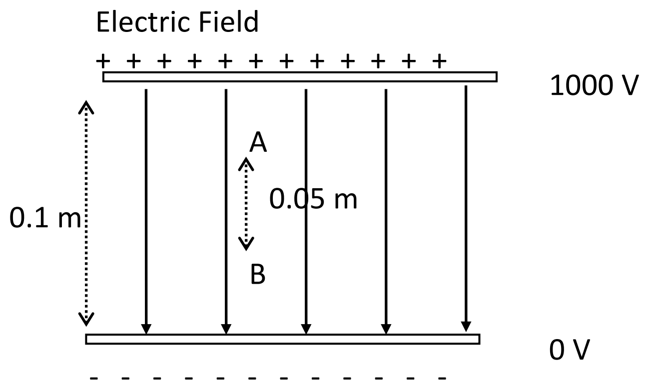
To find the potential difference between A and B, rearrange the equation,

$$E = \frac{\Delta V}{\Delta s}$$

$$\therefore \Delta V = E\Delta s$$

$$\therefore \Delta V = 10000Vm^{-1} \times 0.05m$$

$$\therefore \Delta V = 500V$$



Q=10μC

m=0.1g

Motion in an Electric Field

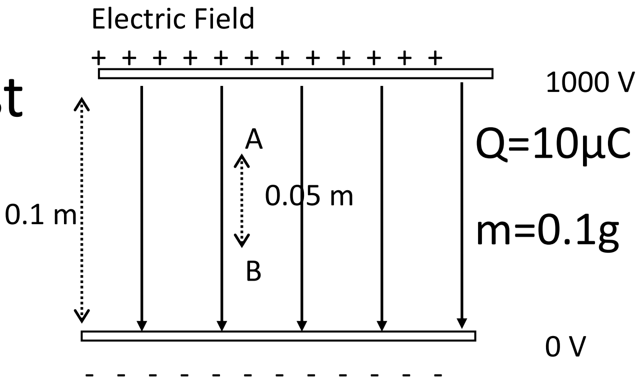
Find the velocity using Change in K.E

(with sample problem)

(Continued)

Now calculate the kinetic energy at point B. If the charge is released at rest,

K.E at B (Gain in *K.E*) = *P.E* lost



$$\tfrac{1}{2}mv^2 = q\Delta V$$

$$\therefore v = \sqrt{\frac{2q\Delta V}{m}}$$

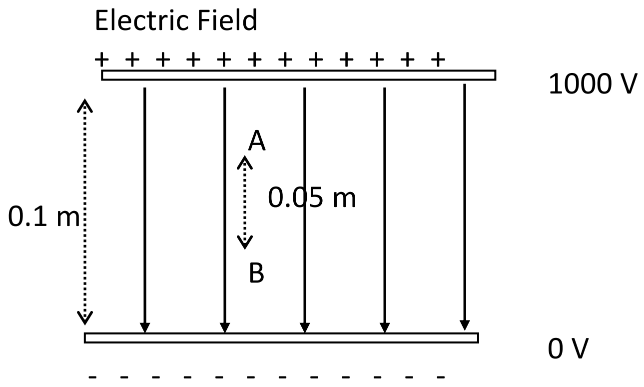
$$= \sqrt{\frac{2\times\left(10\times10^{-6}\right)\times 500}{10^{-4}}}$$

$$= 10ms^{-1} \text{ towards the -'ve plate}$$

Motion in an Electric Field
Find the K.E from the work done
(with sample problem, data from previous slide)

The work done by the field on the charge can be calculated easily because it is equal to the gain in kinetic energy by the charge.

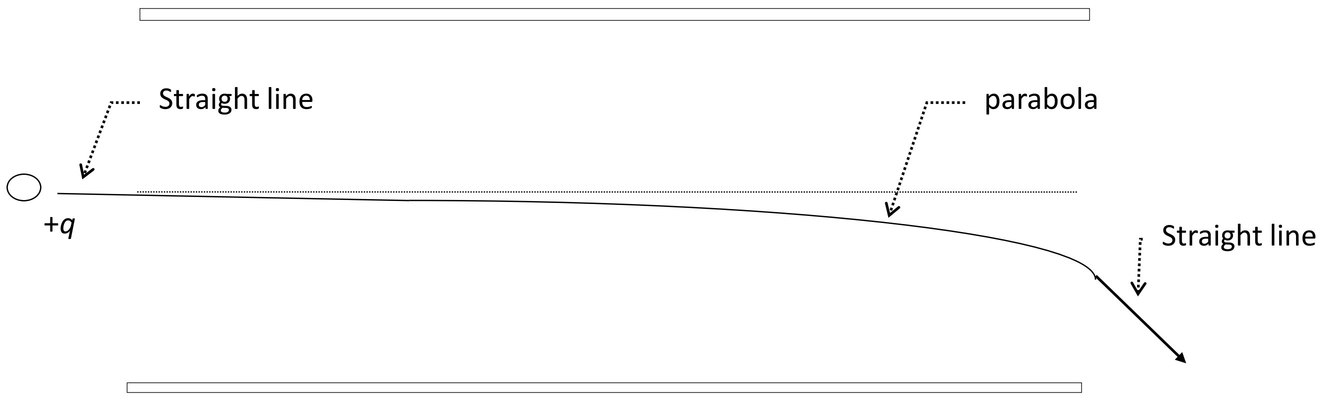
$$\Delta E_K = W = q\Delta V$$
$$= 10^{-5} \times 500$$
$$= 5 \times 10^{-3}$$
$$= 5mJ$$



Q=10μC

m=0.1g

For a charge that enters the Electric Field…



Horizontal Component of the velocity (H component)

$$v_h = v_{1h} = v_{2h} \quad \text{horizontal velocity is constant}$$

$$v_h = \frac{L}{\Delta t} \quad \text{so} \quad \Delta t = \frac{L}{v_h}$$

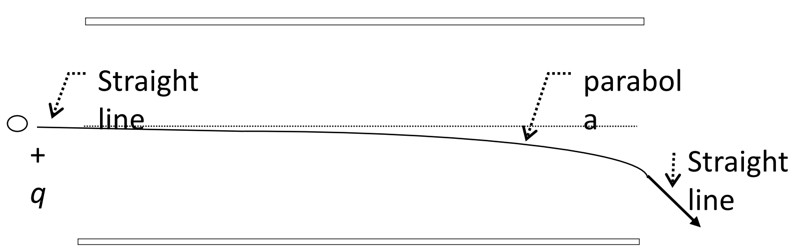
$$a = 0 \quad \text{as} \quad \Delta v = 0$$

For a charge that enters the Electric Field...

(continued from previous slide)

Vertical Component of the velocity (*v* component)

As it is initially travelling **horizontally**,
$v_{y1} = 0$ m/s
Where v_{y1} is initial vertical velocity and v_{y2} is final vertical velocity



$$\text{Now} \quad a_v = \frac{F}{m} = \frac{QE}{m}$$

$$And, \quad \mathrm{v}_{y2} = a\Delta t$$

$$= \frac{QE}{m}\frac{L}{v_x}$$

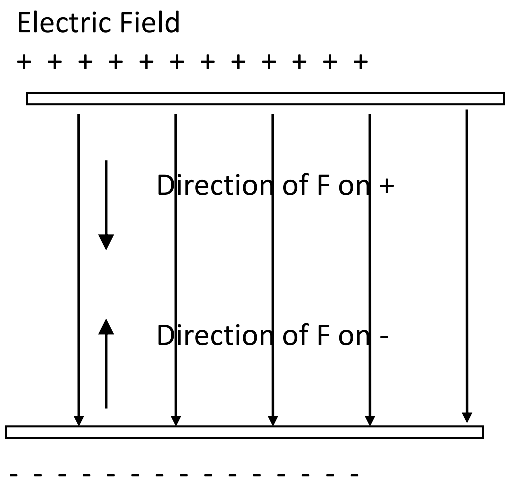
$$So, \quad \Delta \mathrm{s}_v = \tfrac{1}{2} a_v \left(\frac{L}{v_x}\right)^2$$

$$\therefore \Delta s_v = \tfrac{1}{2} a_v \frac{L^2}{v_x^2}$$

v = u + at

S = ut + ½ at

The direction of the force in an electric field



The direction of the force depends on the charge on the particle.

Assumptions
(just for your info only)

Assumptions

- ignore fringe effects (ie. assume that the field is completely uniform)

- ignore gravity (the acceleration due to gravity is insignificant compared with the acceleration caused by the electric field).

> For a charge that enters the field :

-Before entering electric field, the charge follows a **straight line path** (no net force)

-As soon as it enters the field, the charge begins to follow a **parabolic path** (constant force always in the same direction)

- As soon as it leaves the field, the charge follows a **straight line path** (no net force)

Electric field strength (definition)

$$\text{Electric field strength} = \frac{Force}{charge}$$

$$E = \frac{F}{q}$$

$$E = \frac{F}{q}$$

$$W = q\Delta V$$

General definition relationships

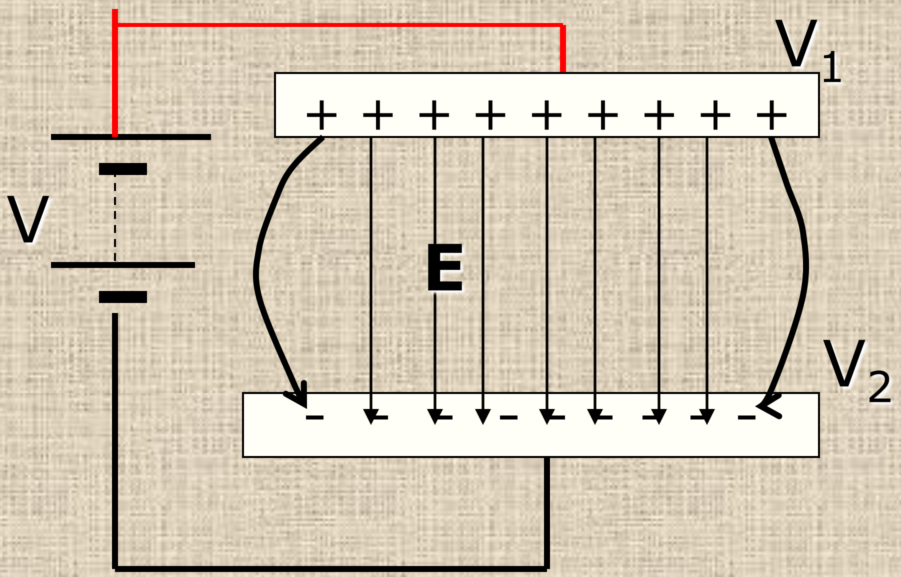
$$E = \frac{V}{d}$$

$$V = Ed$$

Constant field special case relationships

- force per unit charge.
- If the electric field strength is denoted by the symbol **E**, then the equation can be rewritten in symbolic form.
- Unit: N C^{-1} or V m^{-1}

Parallel Conducting Plates - Recall



- Magnitude of electric field strength

$$\boxed{E = \frac{V}{d}}$$

where V = potential difference = $V_1 - V_2$

d = separation between plates

$V_1 > V_2$

- Edge effect. At the edges between the plates the electric field is non-uniform.
- We normally take the electric field between the plates as uniform in calculation and neglect the edge effect.
- **E** points towards lower potential.

Equations of uniform acceleration - Recall

There are four equations for <u>uniform</u> acceleration (also known as the kinematic equations) which are used to describe the motion of an object:

$\mathbf{v} = \mathbf{u} + \mathbf{a}\, t$

$\mathbf{v}^2 = \mathbf{u}^2 + 2\mathbf{as}$

$\mathbf{s} = \frac{1}{2} (\mathbf{u} + \mathbf{v})\, t$

$\mathbf{s} = \mathbf{u}t + \frac{1}{2} \mathbf{a}\, t^2$

Where:

u – <u>initial</u> velocity
v – <u>final</u> velocity
a – <u>acceleration</u>
t – <u>time</u> taken
s – <u>displacement</u>

These are vector equations.

Assumptions

- **<u>acceleration</u>** is constant
- air resistance <u>negligible</u>

17.3 Force between point charges

Coulomb's law

The force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of their distance apart.

Meaning of point charge

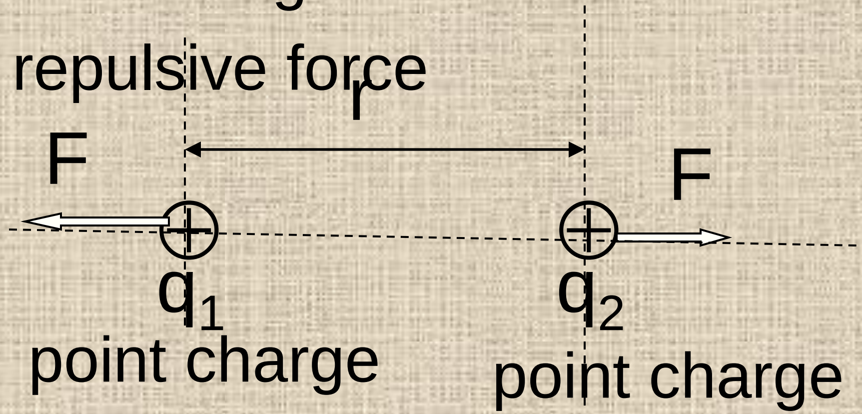
- The separation must be larger than the size of the charge.

$$Electric\ Force = \frac{Qq}{4\pi\varepsilon_o r^2}$$

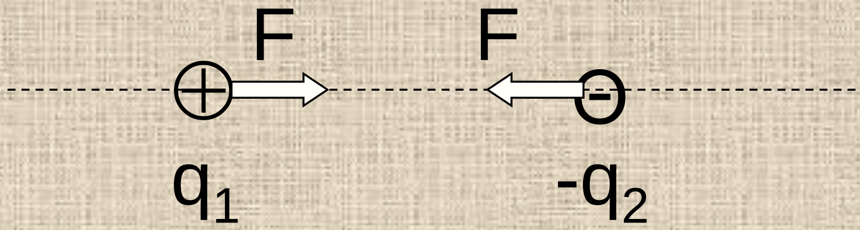
17.3 Force between point charges

Like charges

repulsive force



attractive force



$\varepsilon_o = \frac{q_1 q_2}{4\pi r^2 F}$ Permittivity of free space

- The constant is called the **permittivity of free space.** In SI units where force is in Newtons (N), distance in meters (m) and charge in coulombs (C),
- $\varepsilon_o = 8.85419 \times 10^{-12} \frac{C^2}{Nm^2}$
- (ε_o pronounce as epsilon nought) measures the ease with which the electric field permeates through the medium concern.
- Measures the inverse of the force between unit charges at unit distance apart.

Coulomb's Law

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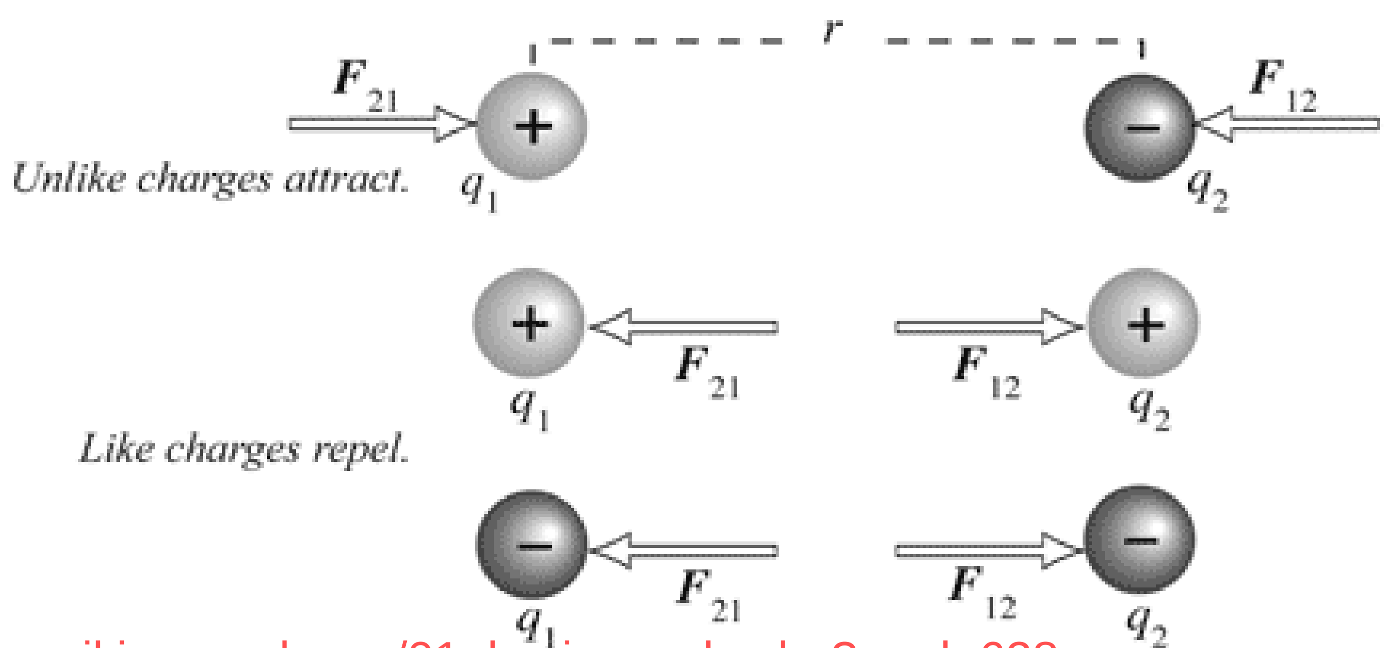
Electrostatic Force - Coulomb's Law

$$k = \frac{1}{4\pi\varepsilon_0}$$

$$F = k\frac{q_1 q_2}{r^2}$$

F = electrostatic force
q = electric charge
r = distance between charge centers
k = Coulomb constant
9.0 X 10^9 N · m²/C²



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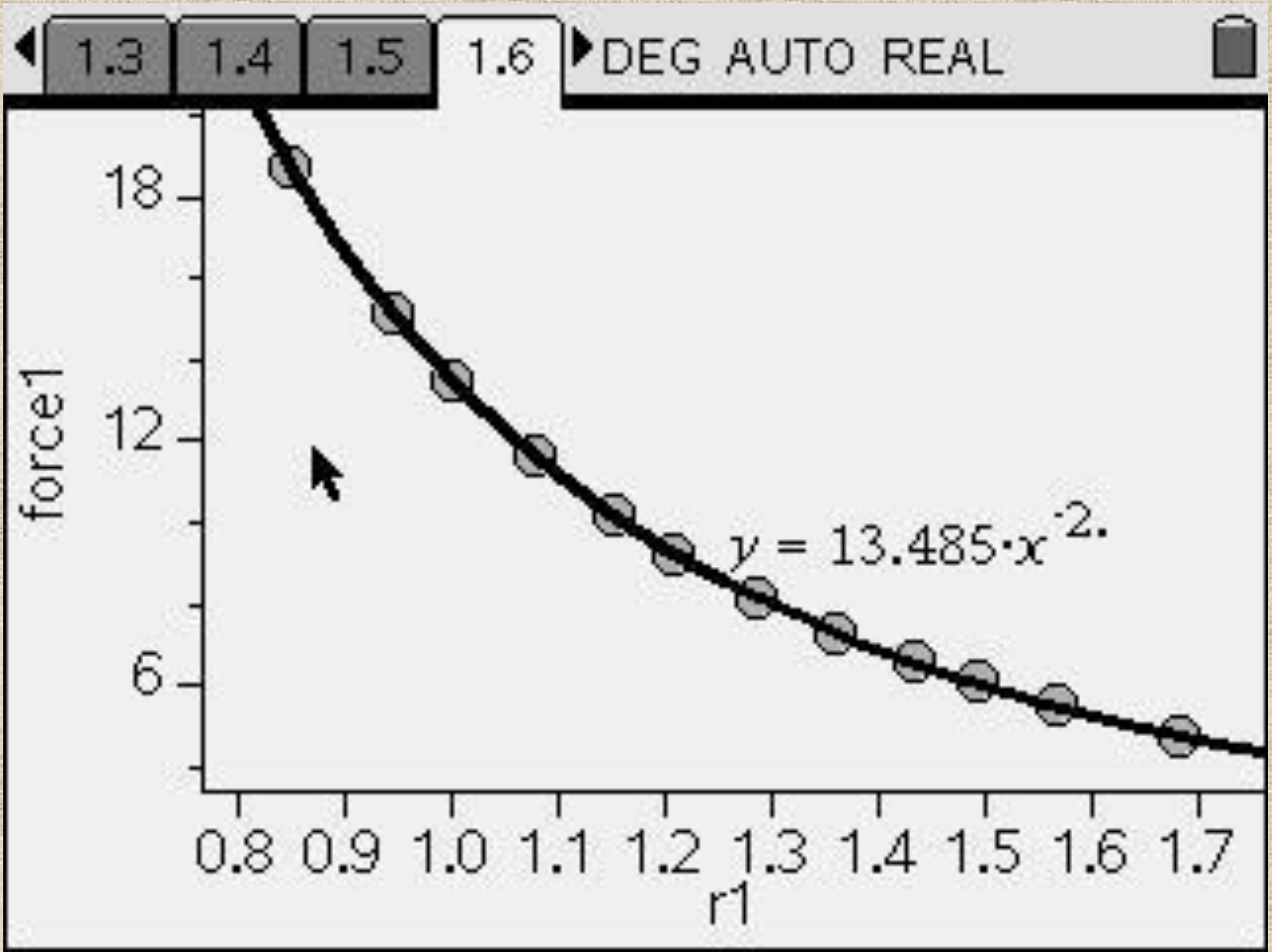
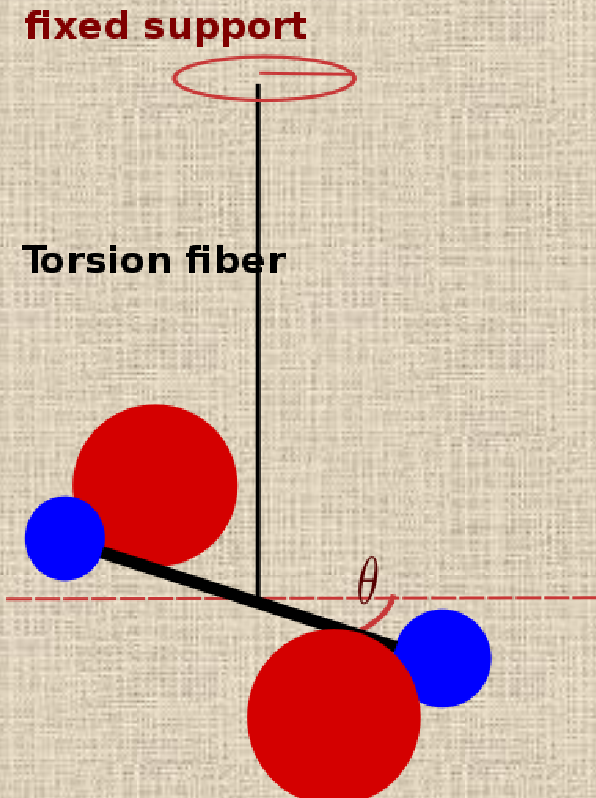
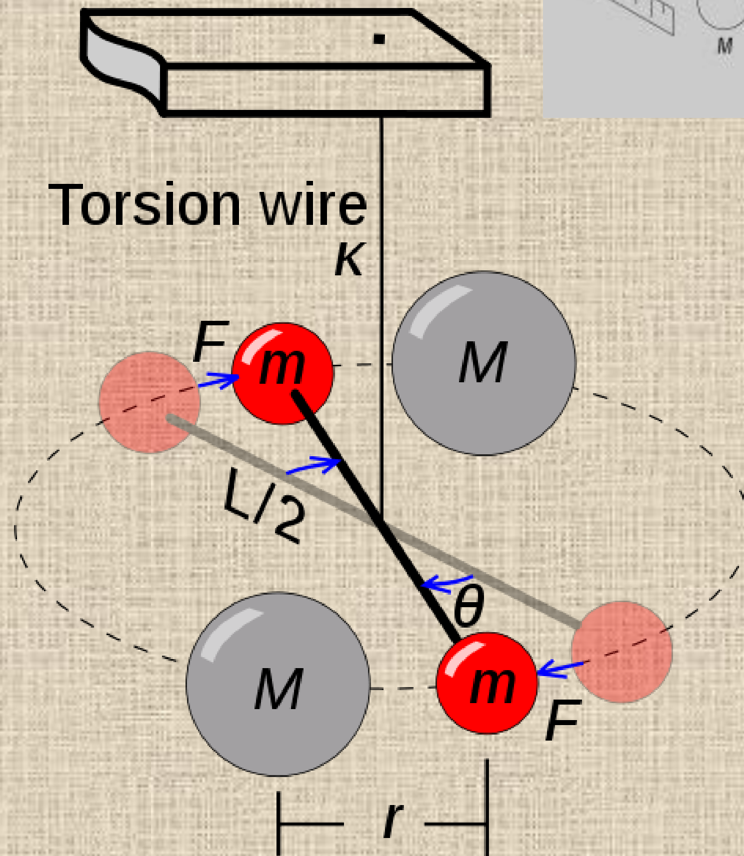
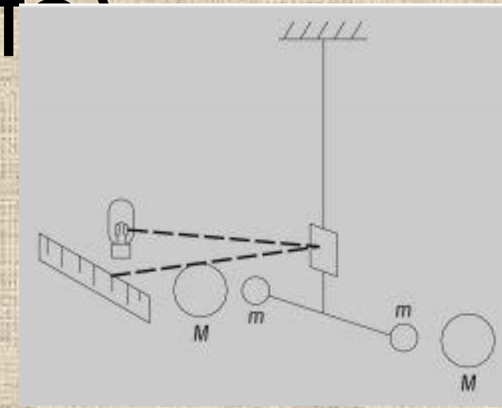
Force between charges as separation increases



Torsion balance (info)







Electric Force

$$F_E = \frac{k\, q_1 q_2}{r^2}$$

Example 17.5

Two charges of 8.0 mC and - 6.0 mC attract each other with a force of 3.0 x 10^3 N in vacuum. Calculate the distance between them.

(Ans. 12 m)

Solution

$$-3x10^3 = 9x10^9 \frac{8x10^{-3}(-6x10^{-3})}{r^2}$$

r = 12 m

[attractive force then it is negative sign]

Example 17.6

Find the force of repulsion between two small positive charges 8.0 nC and 40 nC respectively at a distance apart in air of 10.0 cm.

(Ans 2.9 x 10^{-4})

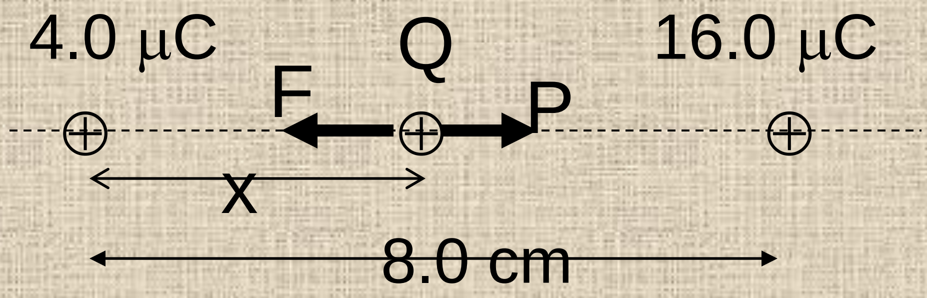
Solution

$$F = 9x10^9\, x\, \frac{8x10^{-9}(40x10^{-9})}{0.1^2}$$

= 2.9x10^{-4} N [repulsive]

Example 17.7

Two charges 4.0 μC and 16.0 μC are separated by a distance of 8.0 cm in vacuum. Find a point where the net force on a charge is zero from the 4.0 μC charge. (Ans. 2.67 cm)



Solution

F = force of 16 μC on Q

$$= \frac{k16Q}{(8-x)^2}$$

P = force of 4 μC on Q

$$= \frac{k4(Q)}{x^2}$$

For net force equals zero,

F= P

$16x^2 = 4(8-x)^2$

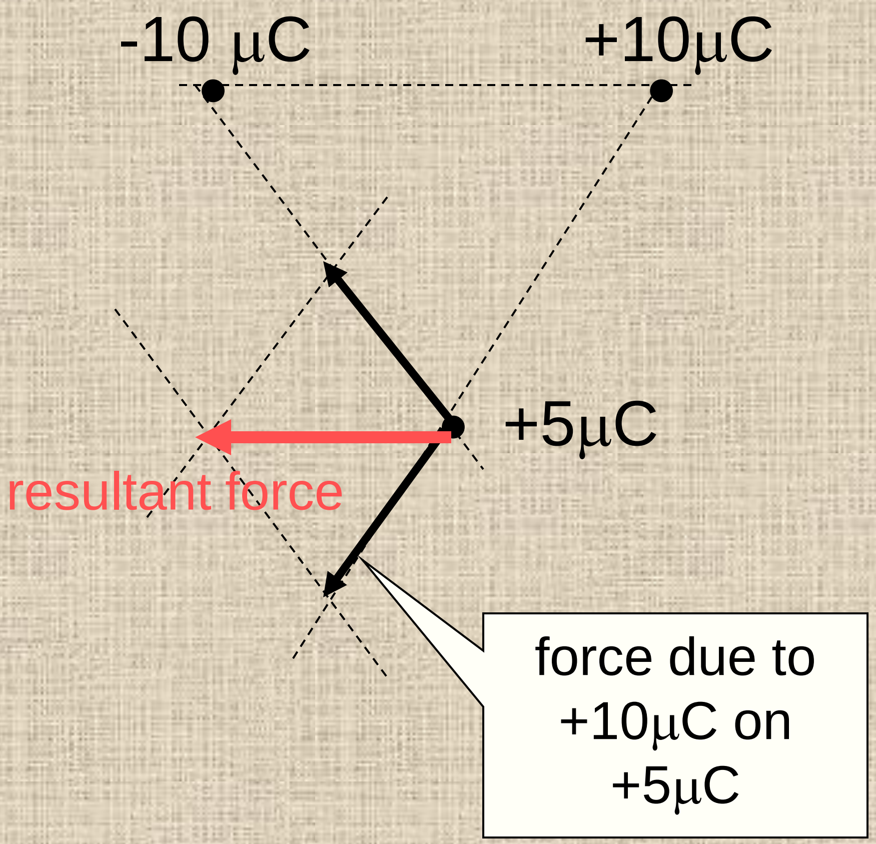
Taking square root,

$4x = \pm 2(8-x)$

x= 2.67 cm

or – 8 cm (impossible)

Example 17.8

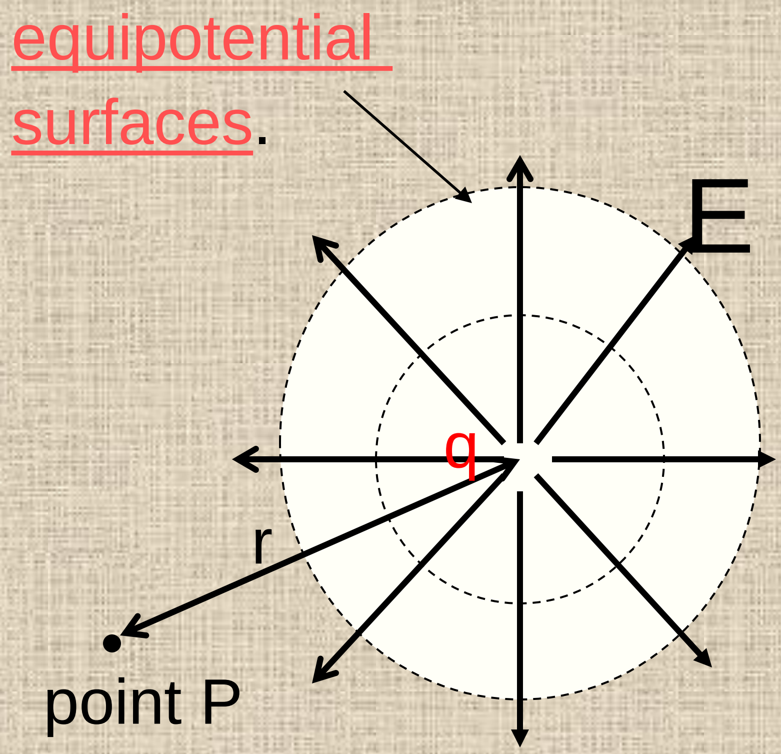


Three point charges are placed at the corner of an equilateral triangle. Draw lines to represent force vectors to show the resultant force acting on 5 μC charge.

17.4 Electric field of a point charge

$$F = E/q$$

$$E = \frac{Q}{4\pi\varepsilon_o r^2}$$



- Non-uniform field.
- field strength decreases with distance from the charge.

Electric field of a point charge q at a Point P a distance r from the charge.

Electric Field and Potential

- The electric field is radially outward from the point charge in all directions. The circles represent spherical equipotential surfaces.
- The electric field lines are at right angle to the equipotential surfaces.
- The electric field from any number of point charges can be obtained from a vector sum of the individual fields. A positive number is taken to be an outward field; the field of a negative charge is toward it.

Derivation of $E = \frac{Q}{4\pi\varepsilon_o r^2}$

Show that the electric field strength due to a point charge q a distance r away is given by

$$E = \frac{Q}{4\pi\varepsilon_o r^2}$$

Steps:

Consider a charge Q establishing an electric field around itself.

Suppose a test charge q is placed a distance r from charge Q.

Force on charge q,

$$F = \frac{Qq}{4\pi\varepsilon_o r^2}$$

Electric field strength = F/q (definition)

$$E = \frac{Q}{4\pi\varepsilon_o r^2}$$

Example 17.11

Calculate the electric field strength at a distance of 8.0 mm from a 40 μC charged particle.

(Ans. 5.62×10^{9} N C^{-1})

Solution

$$E = \frac{q}{4\pi\varepsilon_o r^2}$$

$$= \frac{40x10^{-6}}{4\pi(8.85x10^{-12})(8x10^{-3})^2}$$

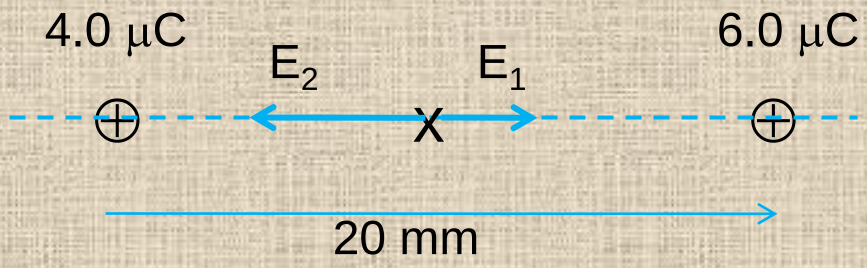
$= 5.62 \times 10^{9}$ N C^{-1}

Example 17.12

Two point charges 4.0 μC and 6.0 μC are 20 mm apart. Calculate the electric field strength at the mid-point between the two charges.

(Ans. 1.8 x10^{8} N C^{-1} towards 4.0 μC)

$$E = \frac{kq}{r^2}$$



Take direction to the right as positive.

Electric field due 4.0 μC

$$E_1 = \frac{kq}{r^2} = \frac{4x10^{-6}}{4\pi(8.85x10^{-12})(0.01)^2}$$

= 3.596 x10^{8} N/C

Electric field due 6.0 μC

$$E_2 = \frac{kq}{r^2} = \frac{-6x10^{-6}}{4\pi(8.85x10^{-12})(0.01)^2}$$

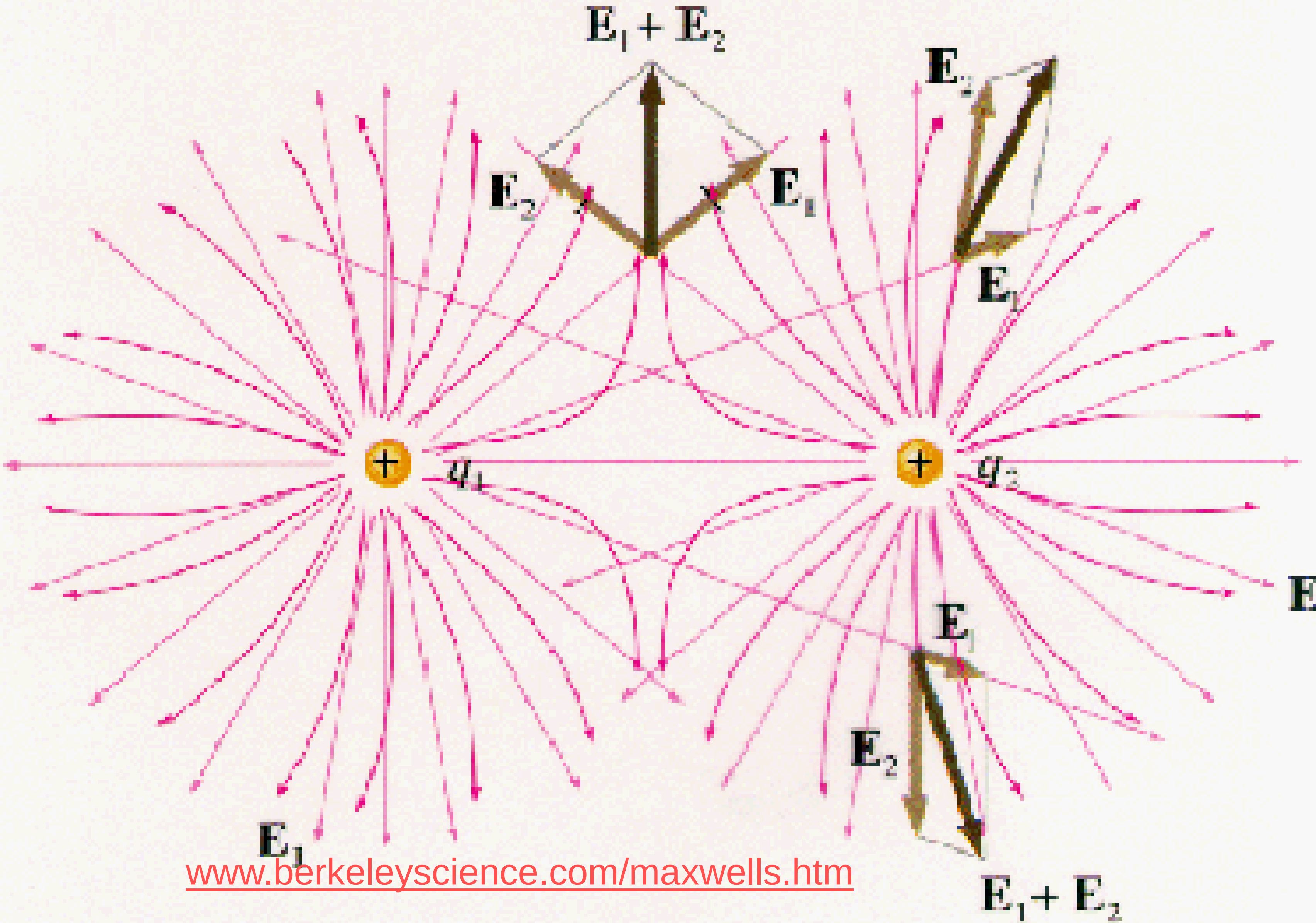
= -5.395 x10^{8} N/C

[negative shows opp. in direction

$E = E_1 + E_2$

= - 1.8x10^{8} N/C

Two Fields E_1 and E_2 Superimpose



www.berkeleyscience.com/maxwells.htm

Electric field and potential

Quantity	New quantity
Force F	electric field strength = force acting on unit charge. $E = F/Q$ E
Work W	Potential = work per unit charge, $V = W/Q$ V
work = F x ***x*** or F = rate of change in energy with distance. $F = dW/dx$	potential = E x ***x*** or E = rate of change in potential with distance $E = - dV/dx$

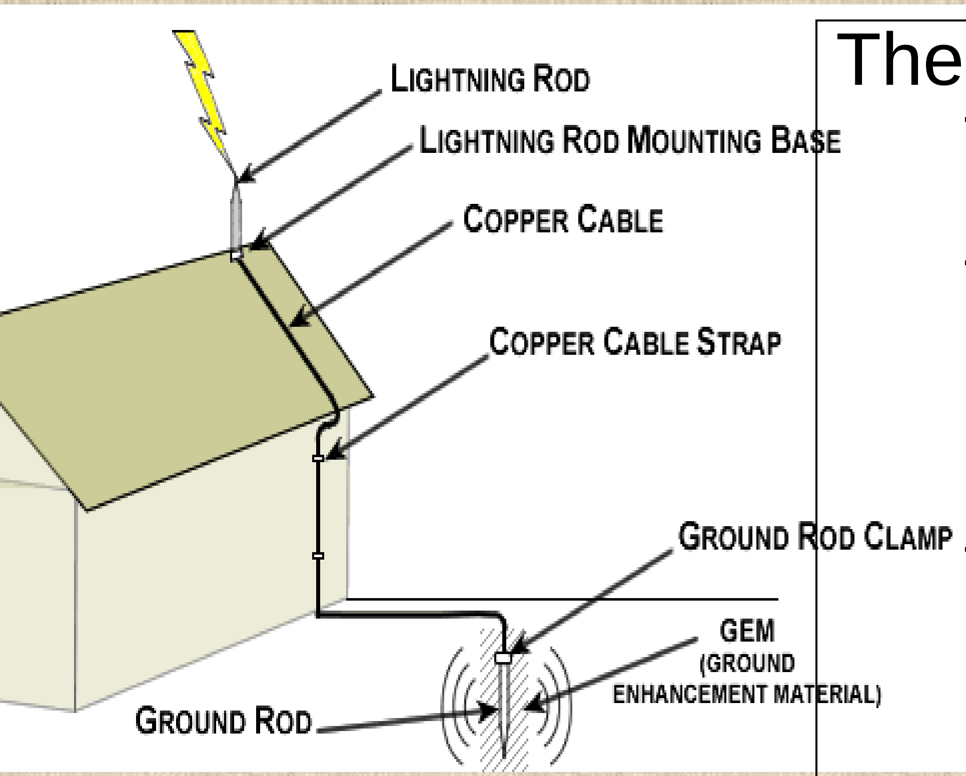
Direction of field

- A test positive charge place near a fixed positive charge is repelled i.e. it accelerate to a place where the resultant force is zero.
- Or we can say that the fixed positive charge creates en electric field round itself. When a test positive charge is place in the electric field it accelerate in the direction of the field to a place where the field is zero.

Action at points.

The point of a pin is highly curved i.e. small radius of curvature, and if it is charged an intense electric field arises near the point. Background ionizing radiation produces electrons in the air which are accelerated by the intense field and cause ionization of the air by collision. Ions having the same sign as the charge of the pin are strongly repelled from it to create an electric ‘wind’. Ions with charges opposite to the pin are attracted to the point and neutralize its charge. The net result of the ‘action at points’ is the apparent loss of charge from the pin to the surrounding air by what is termed a ***point or corona discharge.***

Lightning conductor.



The safe discharge of a thunder cloud by a lightning conductor on a tall building depends on action at points. It occurs at the tip of the metal spike (at the top of the building) which is connected by a thick copper conductor to a plate in the ground. As well as losing charge a sharp point can also collect it.

17.4 Electric potential

- Field strength is a vector and addition by the parallelogram law is more complex.
- When describing a field, potential is usually a more useful quantity than field strength because, it is a scalar and can be added directly when more than one field is concerned.
- When a test positive charge is place near a fixed point charge it experiences a force that accelerates it to infinity (far away) where the force between the charges are zero.

Electric Potential Energy

- We can also think in terms of potential energy. When the test positive charge is place near a fixed charge it had high potential energy, thus it would want to move to a place of zero potential energy where the force is zero.
- The ***potential difference*** (V, or ΔV) between two points in an electric field is the work done in moving a point positive charge, q from one point to the other. Usually from a higher to a lower potential.

  $V = W/q$
- When the point positive charge moved from a higher potential to a lower potential the work done by the field can be thought as a transformation of electric potential energy to a gain in kinetic energy.

Definition.

The ***potential*** at a point is the work done per unit charge in bringing a positive charge from infinity to the point.

- Potential at infinity is zero ($V_\infty = 0$). The force on a charge far away from another charge is zero.
- To calculate the potential at a point we first calculate the work done in moving a point positive charge, q from infinity to the point.
- W = $\int_{\infty}^{r} - F.dr$

Derivation: (for info)

- Suppose the potential V due to a charge Q at P a distance r away is required. If the charge Q creates an electric field around itself, then the force on a point positive charge q place at point P a distance r from the charge is

$$F = k\frac{Q_1Q_2}{r^2} = \frac{Q_1Q_2}{4\pi\varepsilon_o r^2}$$

- the work done on the charge q in bringing it from point P to infinity is $W = \int_r^\infty F.dr \quad = kqQ\int_r^\infty \frac{1}{r^2}\,dr$

$$= kqQ\left[\frac{-1}{r}\right]_r^\infty \quad = \frac{kQq}{r}$$

- Potential at P is

  V = W/q (definition)

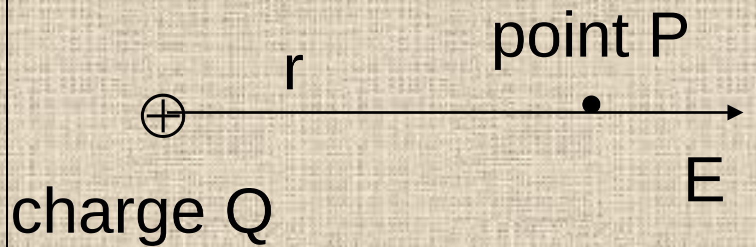
$$V = \frac{Q}{4\pi\varepsilon_o r}$$

Potential at a point

Potential V at a point P a distance r from a point positive charge Q is (no derivation but know how to use).

$$V = \frac{Q}{4\pi\varepsilon_o r}$$

- Potential is more useful than potential energy. It is the potential energy per unit coulomb of test positive charge place in an electric field.
- $E_p = Vq$

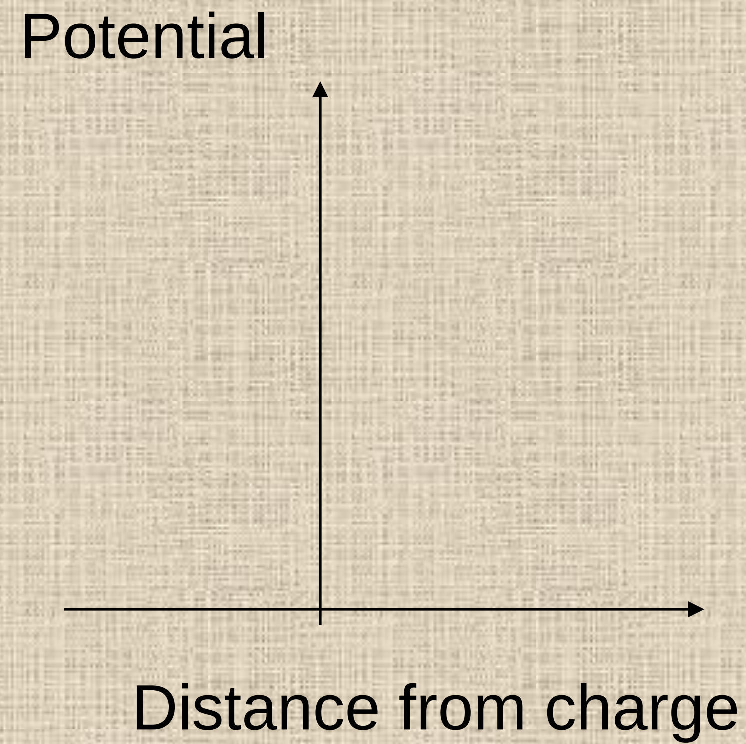


Example 17.14

Sketch the graph of potential V at a point at a distance r from a point positive charge.

Question: What does the gradient of tangent to graph represent?

It is the electric field strength i.e. electric field strength equals potential gradient.



$$E = -\frac{dV}{dr}$$

Potential

- A positive charge is at a high potential as V is positive.
- The potential near a positive point charge is higher than further away.
- Another positive charge place near the charge Q would experience a repulsive force or accelerated away.
- Another way of thinking is to say that the positive charge moves from a high to a low potential along an electric field line.
- Point at equal distance from the charge are at the same potential

Potential

- ***Equipotential surface*** is a surface where the potential is constant. No work is done moving on the surface at constant speed.
- The equipotential surfaces of a point positive charge are spherical surfaces with the charge at the centre.
- The potential at infinity is zero.
- The electric field line is the direction of acceleration of a positive test charge. It is always perpendicular to the equipotential surfaces and points towards lower potential.

Complete the sentences:

A negative point charge place in the electric field would experience an attractive force. It would thus move towards the positive charge along an electric field lines. We can think of this as a negative charge moves from a lower to a higher potential.

True or false? Correct the false statement.

1. The potential due to a point charge is directly proportional to the distance from the charge.
2. If the potential of a point charge at a distance r away is V then the potential at tripled the distance is 3V.
3. The potential at a distance of 0.10 mm from a point charge of 50 μC is 0.5 V.

1) False, potential is inversely proportional to the distance from the charge.
2) False, Since V.r= constant, then the potential is [1/3] V.
3) False

$$V = \frac{Q}{4\pi\varepsilon_o r}$$

$$= \frac{50x10^{-6}}{4\pi(8.85x10^{-12})(0.1x10^{-3})}$$

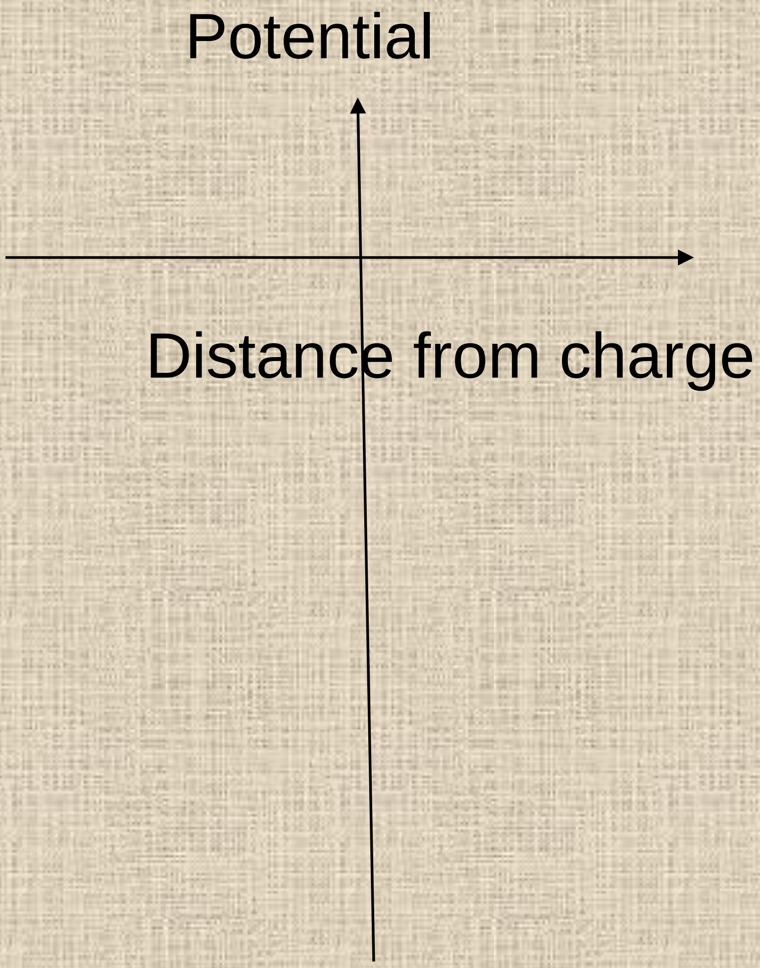
= $4.5x10^{9}$ V

Negative charge

Potential due to a point negative charge –Q at a distance r away.

$$V = -\frac{Q}{4\pi\varepsilon_o r}$$

- Sketch the graph of potential at a point a distance from the negative charge.



Fill in the blank.

- A negative charge is at a low potential.
- Another negative point charge (test charge) places in the field of this negative charge, would move to a higher potential.
- The potential becomes lower as one approaches the negative fixed charge, i.e. potential decreases as distance decreases.

Example 17.16

Triangle ABC is a right-angled triangle with angle C at 90°. If AC = 8.0 cm and AB = 10.0 cm, find,

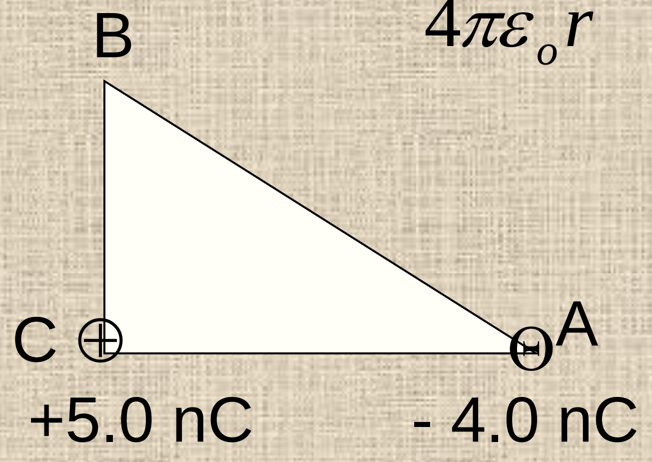
a) the potential at B due to the charge at A,

b) the potential at B due to the charge at C

c) the potential at B due to both charges.

(Ans. a) -360 V b) 750 V c) 390 V)

$$V = \frac{Q}{4\pi\varepsilon_o r}$$



Solution

a) the potential at B due to the charge at A,

$$V_A = \frac{Q}{4\pi\varepsilon_o r} = \frac{-4x10^{-9}}{4\pi(8.85x10^{-12})(0.1)}$$

= -360 V

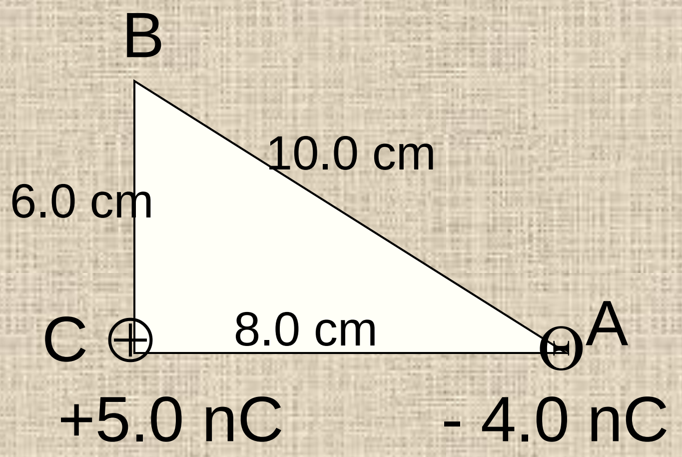
b) the potential at B due to the charge at C

$$V_C = \frac{Q}{4\pi\varepsilon_o r} = \frac{5x10^{-9}}{4\pi(8.85x10^{-12})(0.06)}$$

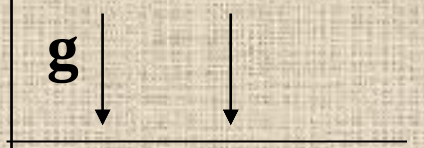
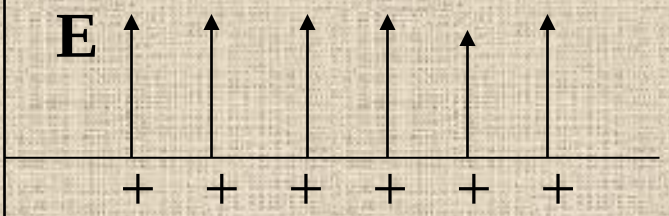
= 750 V

a) the potential at B due to both charges = -360 + 750

= 390 V



Gravitational and Electric fields

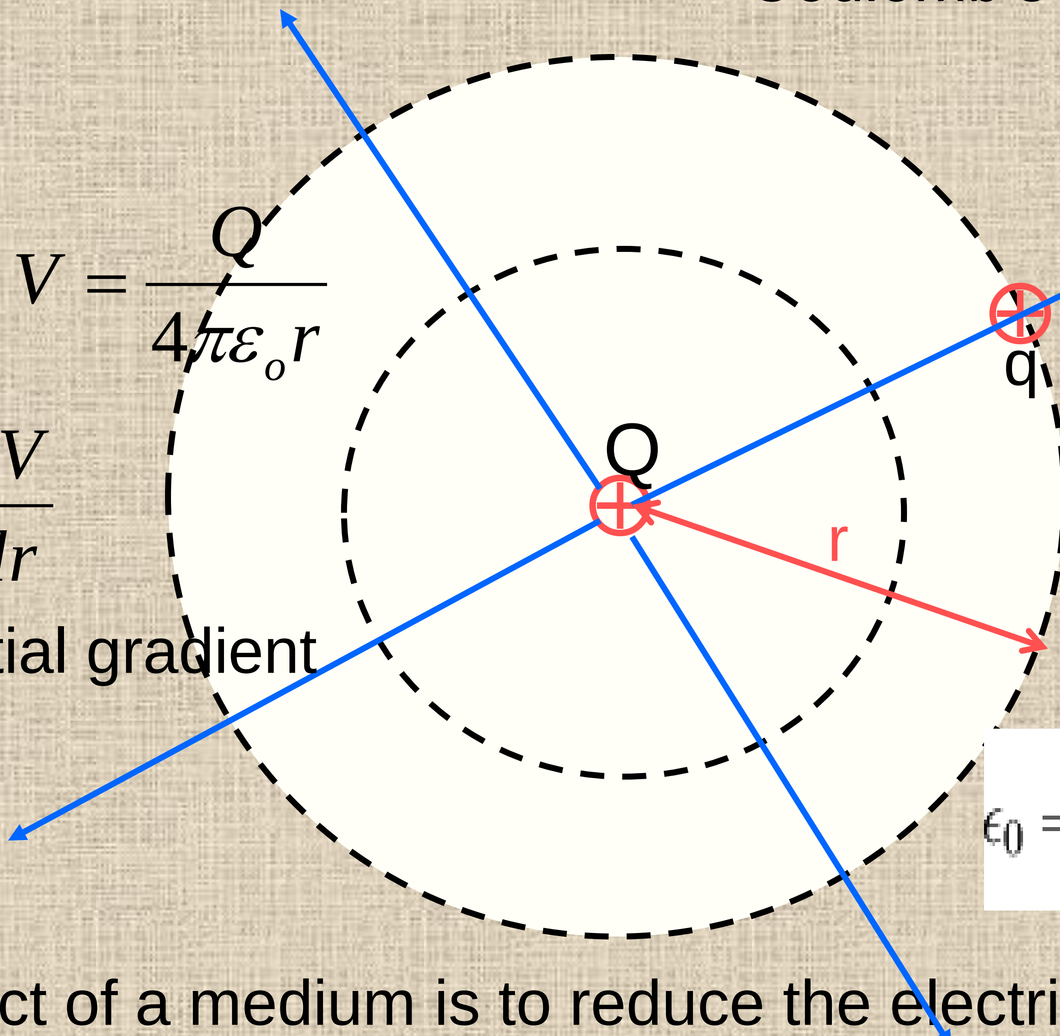
	Gravitational	**Electric**
Field created by	A mass M	A charge Q (can be positive or negative)
Strength	$\mathbf{g} = {}^{F}/m$ force on test mass m	$\mathbf{E} = {}^{\mathbf{F}}/q$ Force on test charge q.
Direction	Towards the mass  Earth	Away from the charge.  Infinitely long plane charged conductor.
Field	attractive	Repulsive on positive charge and attractive on negative charge

Electric field

Coulomb's law $F = \dfrac{Qq}{4\pi\varepsilon_o r^2}$

$$V = \frac{Q}{4\pi\varepsilon_o r}$$

$$E = \frac{kQ}{r^2}$$



$$E = -\frac{dV}{dr}$$

Potential gradient

$$k = \frac{1}{4\pi\varepsilon_0}$$

$$\epsilon_0 = 8.85419 \times 10^{-12} \frac{C^2}{(N\dot{m}^2)}$$

Effect of a medium is to reduce the electric force or field (replace ε_o with ε, the permittivity ofmedium).

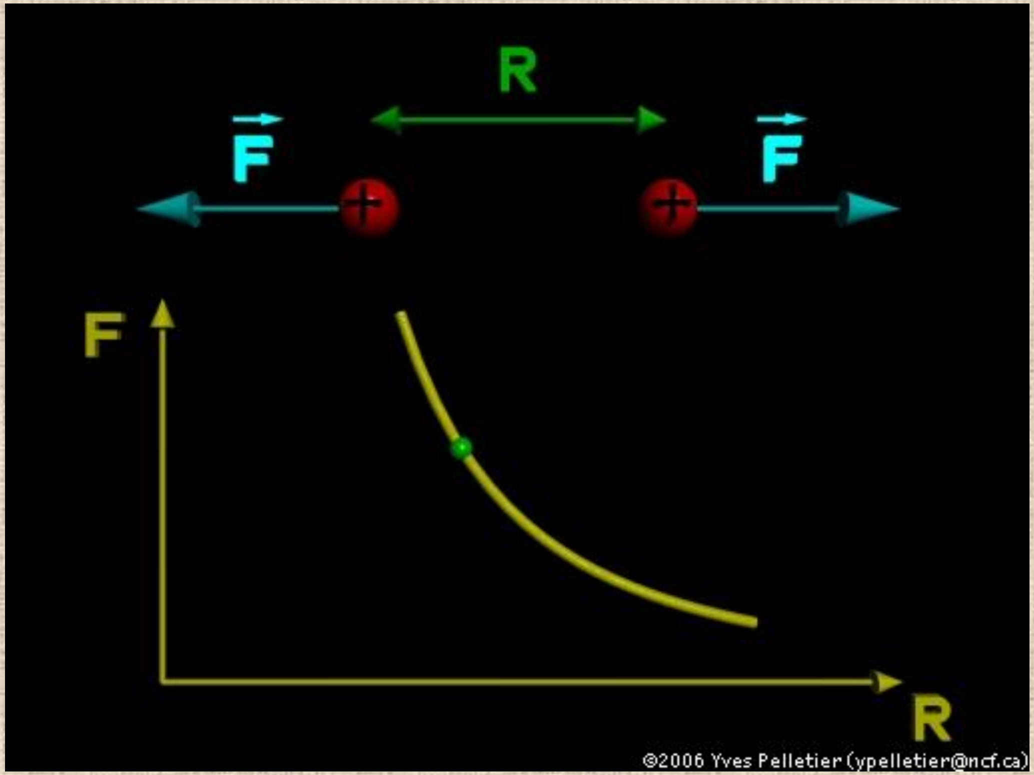
Self test 17.2

1) State Coulomb's law.
2) Define the potential at a point.
3) How is the electric field strength related to the potential gradient?
4) What is the electric field strength and potential of a charge Q at a distance r away?

1) the force of attraction between two point charges is proportional to the product of the charges and inversely proportional to the separation square.
2) Potential at a point is the work done per unit charge in bring the charge from infinity to the point.
3) Electric field strength equals the potential gradient.
4) E and V

$$E = \frac{Q}{4\pi\varepsilon_o r^2} \qquad V = \frac{Q}{4\pi\varepsilon_o r}$$

Info



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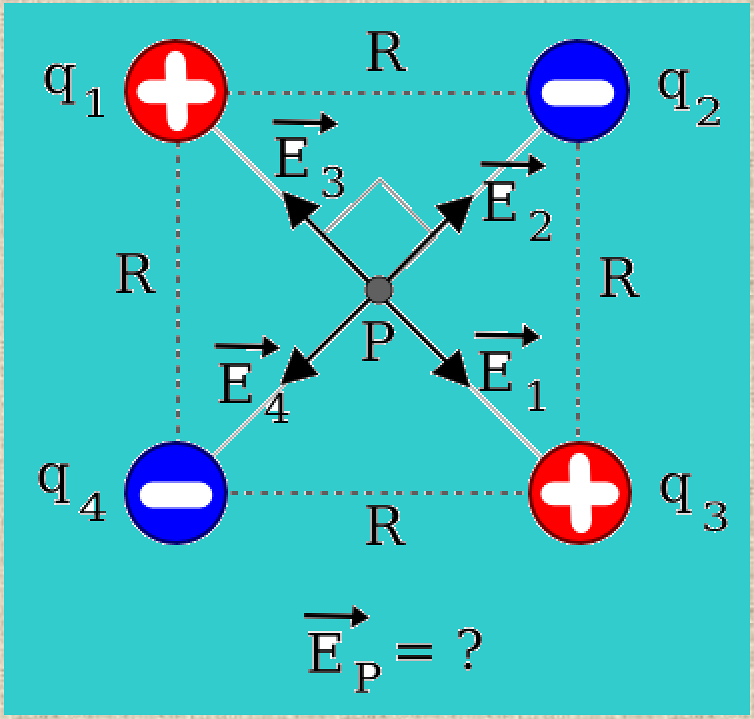

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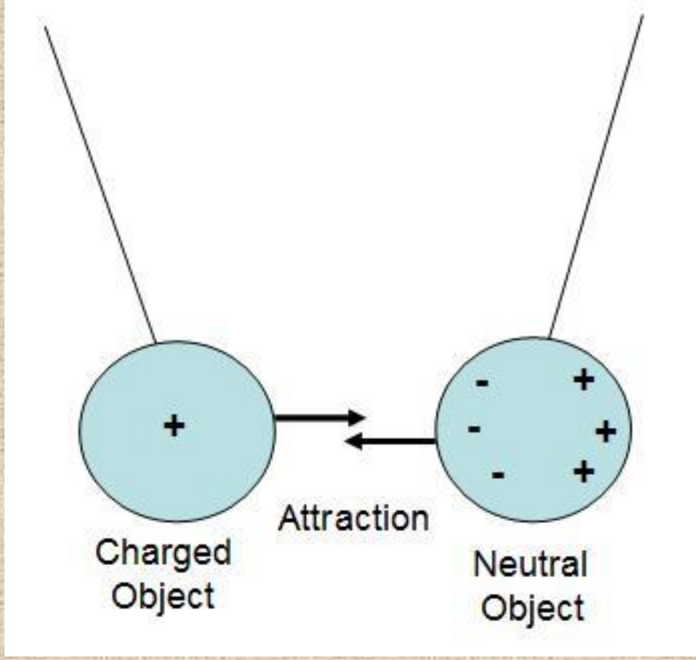
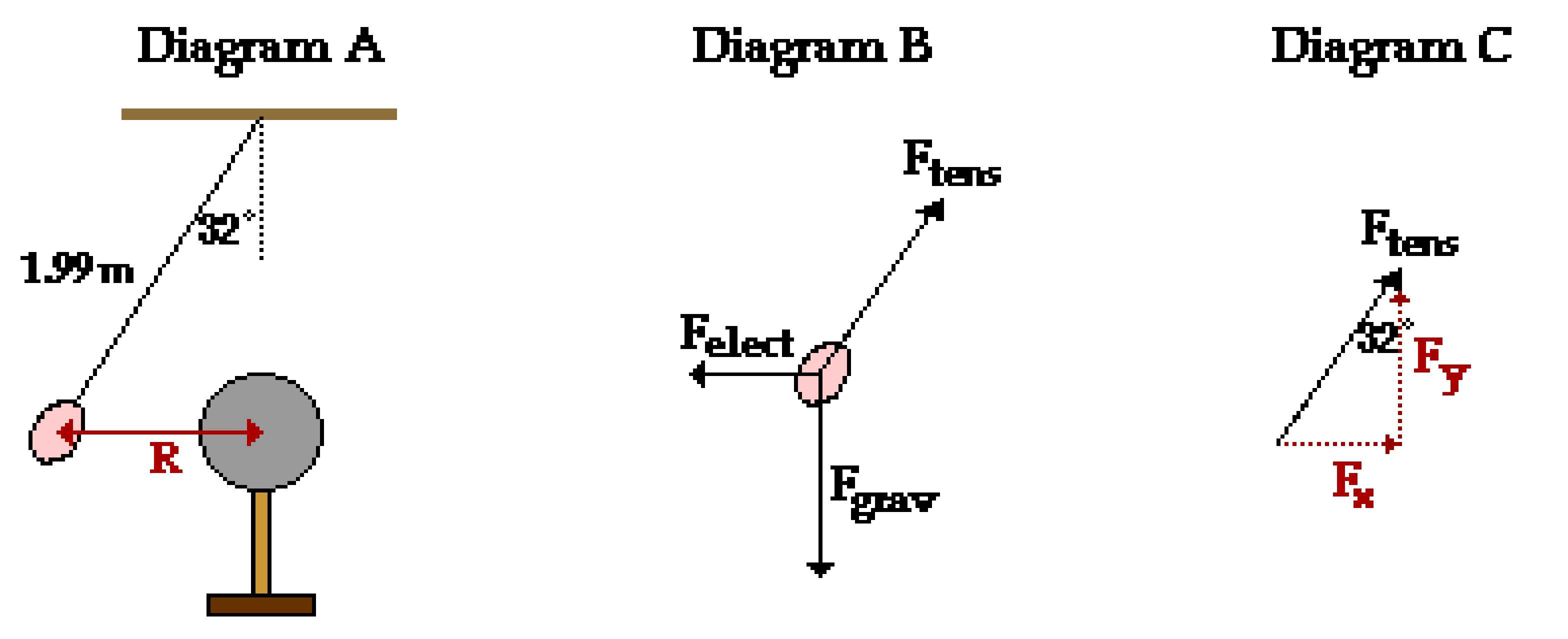


Example 17.17



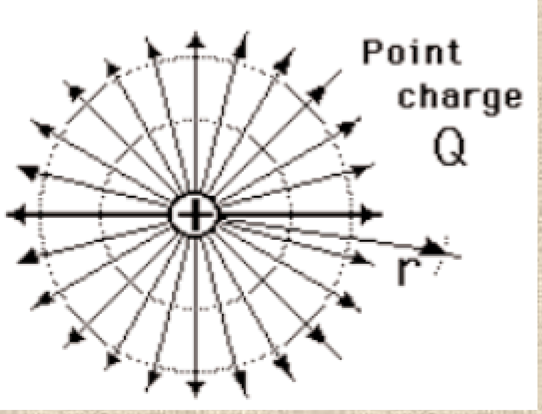
A 1.19 g charged balloon hangs from a 1.99-m string which is attached to the ceiling. A Van de Graaff generator is located directly below the location where the string attaches to the ceiling and is at the same height as the balloon. The string is deflected at an angle of 32^0 from the vertical due to the presence of the electric field. Determine the charge on the Van de Graaff generator if the charge on the balloon is 2.27×10^{-12} C.

Ans. **Q2 = 0.397 C**

gbs.glenbrook.k12.il.us/.../u10ans3.html

Info

$$E(r) = \frac{q}{4\pi\varepsilon_0 r^2}$$



Electric field in N/C or volts/m.

$$\vec{E} = \frac{\vec{F}}{q}$$

electric force in Newtons

charge in Coulombs

Since the measured electric field can depend upon your reference frame, a more general definition of the electric field comes from the Lorentz force law. The electric field can be defined as the electromagnetic force per unit charge in the rest frame of the charge.

$$\vec{F} = \underbrace{q\vec{E}}_{\text{Electric force}} + \underbrace{q\vec{v} \times \vec{B}}_{\text{Magnetic force}}$$

A charge that is moving relative to the source will experience part of the force as a magnetic force.

Lorentz force law

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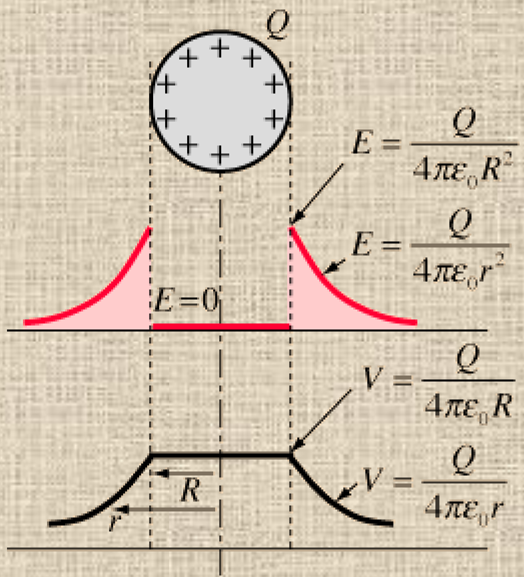

$$V = \frac{kQ}{r} = \frac{Q}{4\pi\varepsilon_0 r}$$

- The use of Gauss' law to examine the electric field of a charged sphere shows that the electric field environment outside the sphere is identical to that of a point charge. Therefore the potential is the same as that of a point charge:
- The electric field inside a conducting sphere is zero, so the potential remains constant at the value it reaches at the surface:

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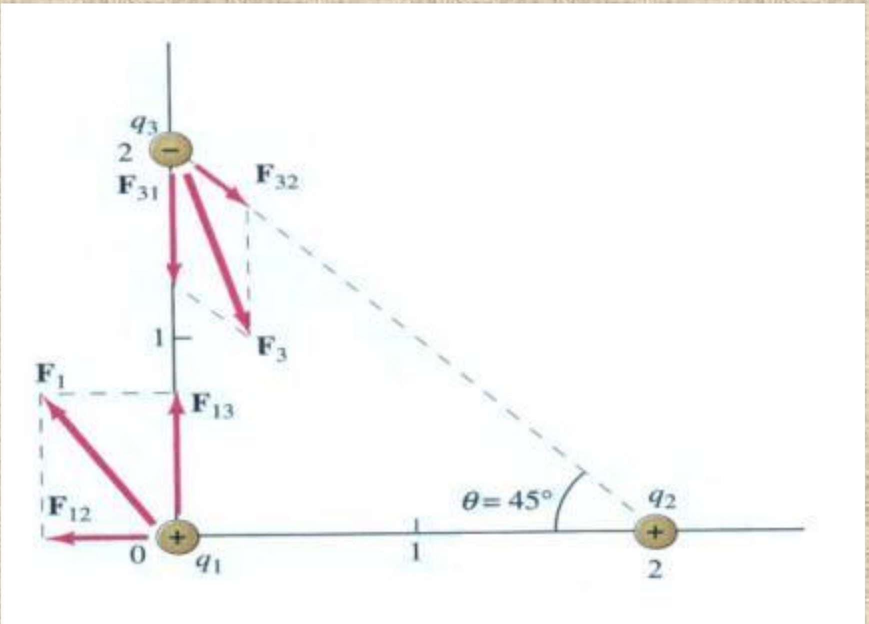


Consider three point charges $q1 = q2 = 2.0$ μC and $q3 = -3$ μC which are placed as shown. Calculate the net force on $q1$ and $q3$.

Ans. 16.2 x10^{-3} N, 56.3º

scitec.uwichill.edu.bb/.../Electric%20fields.htm

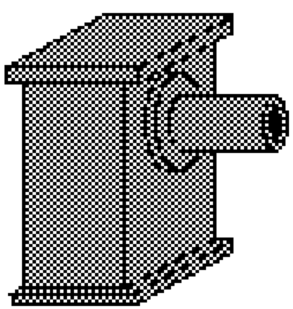
Info

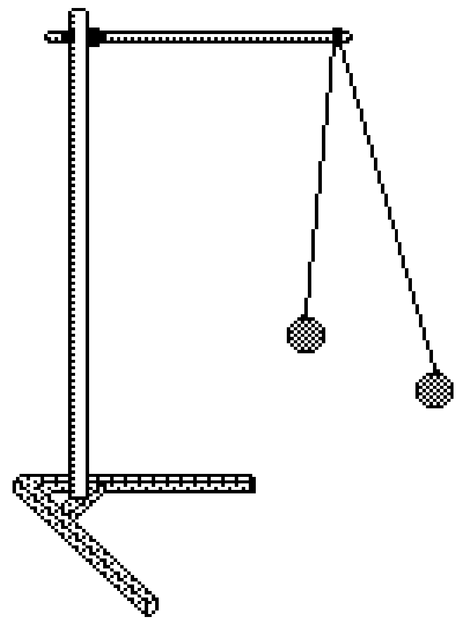
ELECTRICITY AND MAGNETISM | 5A20.20 R | ELECTROSTATICS

Coulomb's Law

COULOMB'S LAW WITH PITH BALLS

- Charge pith balls and measure projected sparation
- Discharge one ball
- Remeasure projected seperation
- The ratio should be 1.59
- Make sure the plane of the balls is parallel to the wall.





R.N.